Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

Normals are drawn at points A, B, and C on the parabola $y^2 = 4x$ which intersect at $P(h, k)$. The locus of the point P if the slope of the line joining the feet of two of them is 2 , is

(A) $x + y = 1$

(B) $x - y = 3$

(C) $y^2 = 2(x - 1)$

(D) $y^2 = 2\left(x - \frac{1}{2}\right)$

Q.2

An equation of the line that passes through $(10, -1)$ and is perpendicular to $y = \frac{x^2}{4} - 2$ is

(A) $4x + y = 39$

(B) $2x + y = 19$

(C) $x + y = 9$

(D) $x + 2y = 8$

Q.3

AB is a focal chord of the parabola $y^2 = 4ax$. If $A = \left(at^2, 2at\right)$, then the area of the quadrilateral formed by the tangents and normals at A and B is

(A) $\frac{a^2}{2}\left(t + \frac{1}{t}\right)$

(B) $2a^2\left(t + \frac{1}{t}\right)$

(C) $a^2\left(t + \frac{1}{t}\right)^2$

(D) $a^2\left(t + \frac{1}{t}\right)^3$

Q.4

If the curves $\frac{x^2}{4} + y^2 = 1$ and $\frac{x^2}{a^2} + y^2 = 1$ for suitable value of 'a' cut on four concyclic points, the equation of the circle passing through these four points is -

(A) $x^2 + y^2 = 2$

(B) $x^2 + y^2 = 1$

(C) $x^2 + y^2 = 4$

(D) None of these

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Q.5

If tangents PQ and PR are drawn from a point on the circle $x^2 + y^2 = 25$ to the ellipse $\frac{x^2}{16} + \frac{y^2}{b^2} = 1$, (b < 4), so that the fourth vertex S of parallelogram PQSR lies on the circumcircle of triangle PQR, then eccentricity of the ellipse is -

(A) $\frac{\sqrt{5}}{4}$

(B) $\frac{\sqrt{7}}{3}$

(C) $\frac{\sqrt{7}}{4}$

(D) $\frac{\sqrt{5}}{3}$

Q.6

If PQ is a double ordinate of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ such that OPQ is an equilateral triangle, O being the centre of the hyperbola. Then the eccentricity e of the hyperbola, satisfies

(A) $e = 1/\sqrt{3}$

(B) $e = 2/\sqrt{3}$

(C) $e = \sqrt{3}/2$

(D) $e > 2/\sqrt{3}$

Q.7

If a hyperbola whose foci are $S \equiv (2, 4)$ and $S' \equiv (8, -2)$ touches x-axis, then equation of hyperbola is

(A) $\frac{(x+y-6)^2}{10} - \frac{(x-y-4)^2}{8} = 1$

(B) $\frac{(x-y-4)^2}{10} - \frac{(x+y-6)^2}{8} = 1$

(C) $\frac{(x-y-4)^2}{20} - \frac{(x+y-6)^2}{16} = 1$

(D) $\frac{(x+y-6)^2}{20} - \frac{(x-y-4)^2}{16} = 1$

Q.8

An ellipse has eccentricity $\frac{1}{2}$ and one focus at the point $\mathrm{P}\left(\frac{1}{2}, 1\right)$. Its one directrix nearer to P is common tangent to the circle $x^2 + y^2 = 1$ and the hyperbola $x^2 - y^2 = 1$, Find the equation of the ellipse.

(A) $9\left(x - \frac{1}{3}\right)^2 + 12(y-1)^2 = 1$

(B) $9\left(x - \frac{1}{3}\right)^2 - 12(y-1)^2 = 1$

(C) $9\left(x - \frac{1}{3}\right)^2 + 8(y-1)^2 = 1$

(D) $12\left(x - \frac{1}{3}\right)^2 + 9(y-1)^2 = 1$

Q.9

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P is a point on the parabola $y^2 = 4x$ where abscissa and ordinate are equal. Equation of a circle passing through the focus and touching the parabola at P is:

(A) $x^2 + y^2 - 13x + 2y + 12 = 0$

(B) $x^2 + y^2 - 3x - 18y + 2 = 0$

(C) $x^2 + y^2 + 13x - 2y - 14 = 0$

(D) $x^2 + y^2 - x = 0$

Q.10

Extremities of the latus rectum of the ellipses $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \quad (a > b)$ having a given major axis 2a lies on

(A) $x^2 = a(a - y)$

(B) $x^2 = a(a + y)$

(C) $y^2 = a(a + x)$

(D) $y^2 = a(a - x)$

Q.11

If $x, y \in R$ then the equation $3x^4 - 2(19y + 8)x^2 + \left(361y^2 + 2\left(100 + y^4\right) + 64\right) = 2\left(190y + 2y^2\right)$ represents in rectangular Cartesian system -

(A) parabola

(B) hyperbola

(C) circle

(D) ellipse

Q.12

Passage

Tangent to the parabola $y = x^2 + ax + 1$, at the point of intersection of y-axis also touches the circle $x^2 + y^2 = r^2$. Also no point of the parabola is below x-axis.

Question:

The radius of circle when a attains its maximum value -

(A) $\frac{1}{\sqrt{10}}$

(B) $\frac{1}{\sqrt{5}}$

(C) 1

(D) $\sqrt{5}$

Q.13

Passage

Tangent to the parabola $y = x^2 + ax + 1$, at the point of intersection of y-axis also touches the circle $x^2 + \ldots$

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$y^2 = r^2$. Also no point of the parabola is below x-axis.

Question:

The slope of the tangent when radius of the circle is maximum is -

(A) -1

(B) 1

(C) 0

(D) 2

Q.14

Passage

Tangent to the parabola $y = x^2 + ax + 1$, at the point of intersection of y-axis also touches the circle $x^2 + y^2 = r^2$. Also no point of the parabola is below x-axis.

Question:

The minimum area bounded by the tangent and the coordinate axes is -

(A) 1

(B) $1/3$

(C) $1/2$

(D) $1/4$

Q.15

Passage

A coplanar beam of light emerging from a point source has the equation $\lambda x - y + 2(1 + \lambda) = 0, \lambda \in R$, the rays of the beam strike an elliptical surface and get reflected. The reflected rays form another convergent beam having equation $\mu \mathrm{x} - \mathrm{y} + 2(1 - \mu) = 0, \mu \in \mathrm{R}$. Further it is found that the foot of the perpendicular from the point $(2, 2)$ upon any tangent to the ellipse lies on the circle $x^2 + y^2 - 4y - 5 = 0$.

Question:

The eccentricity of the ellipse is equal to -

(A) $1/3$

(B) $1/\sqrt{3}$

(C) $2/3$

(D) $1/2$

Q.16

Passage

A coplanar beam of light emerging from a point source has the equation $\lambda x - y + 2(1 + \lambda) = 0, \lambda \in R$, the rays of the beam strike an elliptical surface and get reflected. The reflected rays form another convergent beam having equation $\mu \mathrm{x} - \mathrm{y} + 2(1 - \mu) = 0, \mu \in \mathrm{R}$. Further it is found that the foot of the perpendicular from the point $(2, 2)$ upon any tangent to the ellipse lies on the circle $x^2 + y^2 - 4y - 5 = 0$.

Question:

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The area of the largest triangle that an incident ray and the corresponding reflected ray can enclose with axis of the ellipse is equal to -

(A) $4\sqrt{5}$

(B) $2\sqrt{5}$

(C) $\sqrt{5}$

(D) None of these

Q.17

Passage

A coplanar beam of light emerging from a point source has the equation $\lambda x - y + 2(1+\lambda) = 0, \lambda \in R$, the rays of the beam strike an elliptical surface and get reflected. The reflected rays form another convergent beam having equation $\mu x - y + 2(1-\mu) = 0, \mu \in R$. Further it is found that the foot of the perpendicular from the point $(2, 2)$ upon any tangent to the ellipse lies on the circle $x^2 + y^2 - 4y - 5 = 0$.

Question:

Total distance travelled by an incident ray and the corresponding reflected ray is the least if the point of incidence coincides with -

(A) an end of the minor with

(B) an end of the major axis

(C) an end of this latus rectum

(D) None of these

Q.18

Match the column.

Q.18 Match the column.

	Column I	**Column II**
(A)	Tangents are drawn from point (2, 3) to the parabola $y^2 = 4x$, then points of contact are	(p) (9, – 6)
(B)	From a point P on the circle $x^2 + y^2 = 5$, the equation of chord of contact to the parabola $y^2 = 4x$ is $y = 2(x - 2)$, then the coordinate of point P will be	(q) (1, 2)
(C)	P (4, – 4), Q are point on parabola $y^2 = 4x$ such that area of Δ POQ is 6 sq. units where O is the vertex, then coordinates of Q may be	(r) (–2, 1)
(D)	The common chord of circle $x^2 + y^2 = 5$ and parabola $6y = 5x^2 + 7x$ will pass through point(s)	(s) (4, 4)

(A) A-p,s ; B-q ; C-p,s ; D-r,q

(B) A-q,s ; B-r ; C-p,q ; D-q,r

(C) A-q,s ; B-p ; C-q,p ; D-q,p

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(D) A-s,r ; B-r ; C-r,q ; D-q,r

Q.19

A $(0, 2)$, B and C are points on parabola $y^2 = x + 4$ and such that $\angle$ CBA $= \frac{\pi}{2}$, then find the least positive value of ordinate of C.

Q.20

One side of square $ABCD$ is on the line $y = 2x - 17$ and other two vertices are on parabola $y = x^2$, if the minimum area of square is A then find A/16.

Answers & Solutions

Q.1 Answer:

$x - y = 3$

Solution:

The equation of normal at $(a^2, 2at)$ is

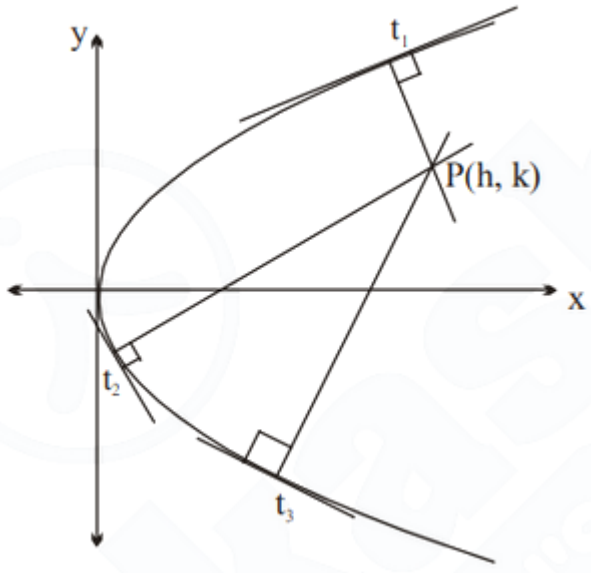
$y + tx = 2at + at^3$ **...(1)**

As (1) passes through $\mathrm{P(h, k)}$, so

$\mathrm{at^3 + t(2a - h) - k = 0} \longleftarrow_{t_2}^{t_2}$ **...(2)**

Here $a = 1$

$t_1 + t_2 + t_3 = 0$ **...(3)**



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Also $\frac{2}{t_1+t_2} = 2 \Rightarrow t_1 + t_2 = 1$ **...(4)**

From (3) and (4) $\Rightarrow t_3 = -1$

Put $t_3 = -1$ **in (2), we get**

$$-1 - 1(2 - h) - k = 0$$
$$\Rightarrow -1 - 2 + h - k = 0$$

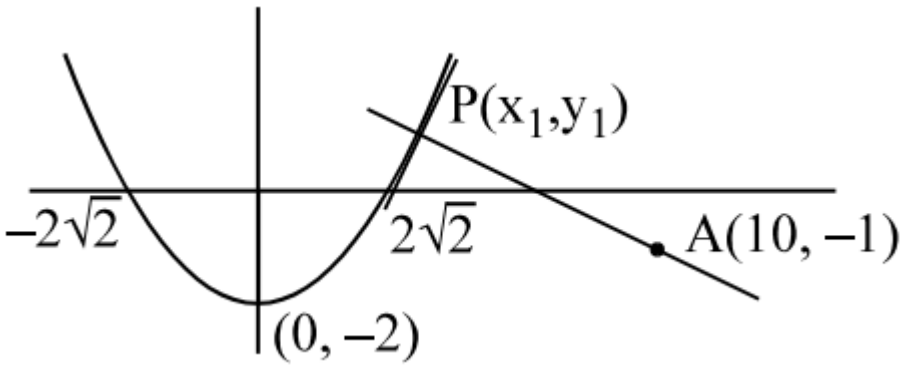
$\therefore$ **Locus of** $P(h, k)$**, is** $x - y = 3$

Q.2 Answer:

$x + 2y = 8$

Solution:

$$4y = x^2 - 8$$
$$4\frac{dy}{dx} = 2x$$
$$\left.\frac{dy}{dx}\right|_{x_1, y_1} = \frac{x_1}{2}$$



$\therefore$ Slope of normal $= -\frac{2}{x_1}$; but slope of normal $= \frac{y_1 + 1}{x_1 - 10}$

$\therefore \quad \frac{y_1 + 1}{x_1 - 10} = -\frac{2}{x_1}$

$\Rightarrow x_1 y_1 + x_1 = -2x_1 + 20$

$\Rightarrow x_1 y_1 + 3x_1 = 20$

substituting $y_1 = \frac{x_1^2 - 8}{4}$ **(from the given equation)**

$$x_1\left(\frac{x_1^2 - 8}{4} + 3\right) = 20$$
$$\Rightarrow x_1\left(x_1^2 - 8 + 12\right) = 80$$
$$\Rightarrow x_1\left(x_1^2 + 4\right) = 80$$
$$x_1^3 + 4x_1 - 80 = 0$$

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$x_1^2(x_1-4)+4x(x_1-4)+20(x_1-4)=0$

$(x_1-4)\left(x_1^2+4x_1+20\right)=0$

Hence $x_1=4; y_1=2$

$\therefore \quad P=(4,2)$

equation of PA **is**

$y+1=-\frac{1}{2}(x-10)$

$\Rightarrow 2y+2=-x+10$

$\Rightarrow x+2y-8=0$

Q.3 Answer:

$a^2\left(t+\frac{1}{t}\right)^3$

Solution:

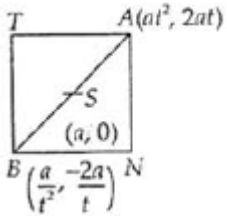
The tangents at A **and** B **meet at** $T=\left(-a, a\left(t-\frac{1}{t}\right)\right)$

The normals at A **and** B **meet at** N**. TANB is a rectangle.**

$TA^2=a^2\left(t^2+1\right)^2+a^2\left(t+\frac{1}{t}\right)^2=\frac{a^2}{t^2}\left(t^2+1\right)^3$

$TB^2=a^2\left(\frac{1}{t^2}+1\right)^2+a^2\left(t+\frac{1}{t}\right)^2=\frac{a^2}{t^4}\left(t^2+1\right)^3$

Area of TANB is TA $\cdot TB=\frac{a^2}{t^3}\left(t^2+1\right)^3=a^2\left(t+\frac{1}{t}\right)^3$.



Q.4 Answer:

$x^2+y^2=1$

Solution:

Equation of conic through point of intersection of given two ellipse is

$\left(\frac{x^2}{4}+y^2-1\right)+\lambda\left(\frac{x^2}{a^2}+y^2-1\right)=0$

$\Rightarrow \mathrm{x}^2\left(\frac{1}{4}+\frac{\lambda}{\mathrm{a}^2}\right)+\mathrm{y}^2(1+\lambda)=1+\lambda \Rightarrow \mathrm{x}^2\left(\frac{\mathrm{a}^2+4\lambda}{4\mathrm{a}^2(1+\lambda)}\right)+\mathrm{y}^2=1$

This equation is circle if $\frac{\mathrm{a}^2+4\lambda}{4\mathrm{a}^2(1+\lambda)}=1 \Rightarrow$ **Circle is** $\mathrm{x}^2+\mathrm{y}^2=1$

Q.5 Answer:

$\frac{\sqrt{7}}{4}$

Solution:

A cyclic parallelogram will be a rectangle or square. So, $\angle \mathrm{QPR}=90^{\circ}$

$\Rightarrow P$ **lies on director circle of the ellipse** $\frac{x^2}{16}+\frac{y^2}{b^2}=1$

$\Rightarrow \quad x^2+y^2=25$ **is director circle of** $\frac{x^2}{16}+\frac{y^2}{b^2}=1 \Rightarrow 16+b^2=25 \Rightarrow b^2=9$

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$\Rightarrow \quad a^2\left(1-e^2\right)=9 \Rightarrow 1-e^2=\frac{9}{16} \Rightarrow e^2=\frac{7}{16}$

$\Rightarrow e=\frac{\sqrt{7}}{4}$

Q.6 Answer:

$e > 2/\sqrt{3}$

Solution:

If OPQ is equilateral triangle then OP makes 30° with x-axis.

$\left(\frac{\sqrt{3}r}{2}, \frac{r}{2}\right)$ **lies on hyperbola** $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$

$$\Rightarrow r^2=\frac{16a^2b^2}{12b^2-4a^2}>0$$

$$\Rightarrow 12b^2-4a^2>0 \Rightarrow \frac{b^2}{a^2}>\frac{4}{12}$$

$$e^2-1>\frac{1}{3}$$

$$e^2>\frac{4}{3} \Rightarrow e>\frac{2}{\sqrt{3}}$$

Q.7 Answer:

$\frac{(x-y-4)^2}{20}-\frac{(x+y-6)^2}{16}=1$

Solution:

$b^2=d_1d_2$, **where** d_1 **and** d_2 **are perpendicular distances of any tangent from focii of the hyperbola**

$b^2=4\times 2=8 \quad \Rightarrow \quad b=2\sqrt{2}$

$2ae=\sqrt{72}=6\sqrt{2}$

So $a^2=10$

Centre is $(5,1)$

Equation of conjugate axis $x-y=4$ **and equation of transverse axis is** $x+y=6$

$\therefore$ **Equation of hyperbola is**

$\frac{(x-y-4)^2}{20}-\frac{(x+y-6)^2}{16}=1$

Q.8 Answer:

$9\left(x-\frac{1}{3}\right)^2+12(y-1)^2=1$

Solution:

The circle $x^2+y^2=1$ **and the hyperbola** $x^2-y^2=1$ **touch each other at the points** $(\pm 1,0)$ **and the common tangent at these point are** $x=\pm 1$.

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Since $x = 1$ is nearer to the focus $P\left(\frac{1}{2}, 1\right)$, this is the directrix of the required ellipse.

Therefore, the major axis is parallel to the axis passing through the focus $P\left(\frac{1}{2}, 1\right)$.

Hence the equation of the major axis is $y = 1$.

Let a be the length of the semi major axis of the ellipse and let the coordinates of the centre C of the ellipse be $(\alpha, 1)$.

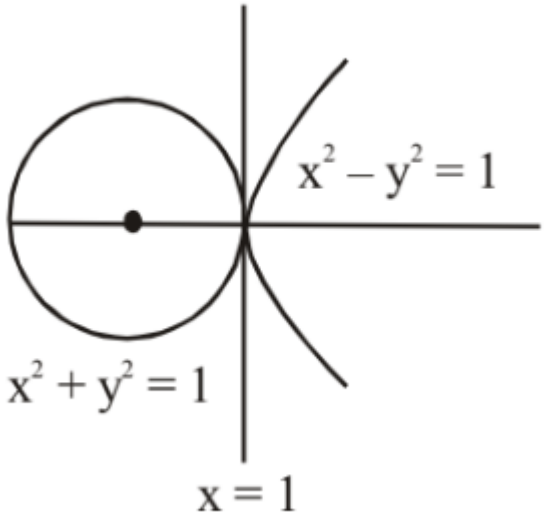
Then $\mathrm{CP} = \frac{1}{2} - \alpha = \mathrm{a.e} = \mathrm{a} \times \frac{1}{2} \ldots$ **(i)** $\left(\because \mathrm{e} = \frac{1}{2}\right)$ **and the distance of the directrix from the centre** $= \mathrm{a/e}$.

$\Rightarrow 1 - \alpha = \frac{\mathrm{a}}{\mathrm{e}} = 2\mathrm{a}$

From (i) and (ii) we get $\mathrm{a} = \frac{1}{3}$ **and** $\mathrm{e} = \frac{1}{2}$.

If b is the length of the semi minor axis of the ellipse, then $b^2 = a^2\left(1 - e^2\right)$

$\Rightarrow \mathrm{b}^2 = \frac{1}{9}\left(1 - \frac{1}{4}\right) = \frac{1}{12}$



Hence the required equation of the ellipse is

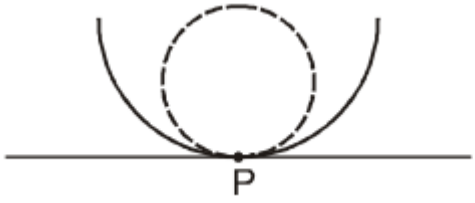
$\frac{\left(x - \frac{1}{3}\right)^2}{(1/3)^2} + \frac{(y-1)^2}{1/12} = 1$ or $9\left(x - \frac{1}{3}\right)^2 + 12(y-1)^2 = 1$

Q.9 Answer:

$x^2 + y^2 - 13x + 2y + 12 = 0$

Solution:

$\mathrm{at}^2 = 2\mathrm{at}$ point



$t = 0, t = 2, (0, 0)$

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(I) when $P \equiv (0,0)$

$x^2 + y^2 + \lambda(x) = 0$

pass the $(1,0)$

$\lambda = -1$, **equation tagent al** $(0,0)$

$y^2 = 4x$

Equ. $x^2 + y^2 - x = 0$

y. $y_1 = 2(x + x_1)$

$x = 0$

(II) when point $(4,4), 2x - 2y + 8 = 0$

$(x-4)^2 + (y-4)^2 + \mu(2x - 2y + 8) = 0$

pass $(1,0)$

Equation

$x^2 + y^2 - 13x + 2y + 12 = 0$

Q.10 Answer:

$x^2 = a(a - y)$

Solution:

$h = \pm ae; k = \pm \frac{b^2}{a}$

$k = \pm a\left(1 - e^2\right) = \pm a\left(1 - \frac{h^2}{a^2}\right) = \pm\left(a - \frac{h^2}{a}\right)$

$+$ ve sign, $k = a - \frac{h^2}{a}$

$\Rightarrow \frac{h^2}{a} = a - k$

$\Rightarrow h^2 = a(a-k) \Rightarrow (A)$

$-ve$ sign, $k = -a + \frac{h^2}{a}$

$\Rightarrow h^2 = a(a+k)$

Q.11 Answer:

parabola

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Solution:

$3x^4 - 2(19y + 8)x^2 + \left[(19y)^2 + (10)^2 + (10)^2 + y^4 + y^4 + 8^2\right] =$
$2\left(19 \times 10y + 10y^2 - 8y^2\right)$
$\Rightarrow 3x^4 - 2(19y + 8)x^2 + (19y - 10)^2 + \left(10 - y^2\right)^2 + \left(y^2 + 8\right)^2 = 0$
$\Rightarrow 3x^4 - 2\left(19y - 10 + 10 - y^2 + y^2 + 8\right)x^2 + (19y - 10)^2 + \left(10 - y^2\right)^2 + \left(y^2 + 8\right)^2 = 0$
$\Rightarrow \quad \left[x^2 - (19y - 10)\right]^2 + \left[x^2 - \left(10 - y^2\right)\right]^2 + \left[x^2 - \left(y^2 + 8\right)\right]^2 = 0$
$\Rightarrow \quad x^2 - 19y + 10 = 0, x^2 - 10 + y^2 = 0$ **and** $x^2 - y^2 - 8 = 0$
The three curves represented by the given equation are $x^2 = 19y - 10$ **(parabola),** $\mathrm{x^2 + y^2 = 10}$ **(circle) and** $\mathrm{x^2 - y^2 = 8}$ **(hyperbola).**

Q.12 Answer:

$\frac{1}{\sqrt{5}}$

Solution:

(i) Since no point of the parabola is below x-axis
$\therefore \quad D = a^2 - 4 \leq 0$
Therefore, maximum value of a is 2 .
Equation of the parabola, when $\mathrm{a = 2}$**, is** $\mathrm{y = x^2 + 2x + 1}$
It intersects y**-axis at** $(0, 1)$**.**
Equation of the tangent at $(0, 1)$ **is** $\quad y = 2x + 1$
Since, $y = 2x + 1$ **touches the circle** $x^2 + y^2 = r^2$
$\therefore \quad \mathrm{r = \frac{1}{\sqrt{5}}}$

Q.13 Answer:

0

Solution:

Equation of the tangent at $(0, 1)$ **to the parabola** $\mathrm{y = x^2 + ax + 1}$ **is**
$\mathrm{or\ y - 1 = a(x - 0)}$
$\mathrm{As - y + 1 = 0}$
$\mathrm{A\ it\ touches\ the\ circle, r = \frac{1}{\sqrt{a^2+1}}}$
Radius is maximum when $\mathrm{a = 0}$
Therefore, equation of the tangent is $\mathrm{y = 1}$
Therefore, slope of the tangent is 0.

Q.14 Answer:

$1/4$

Solution:

(iii) Equation of tangent is $\mathrm{y = ax + 1}$
Intercepts are $-1/\mathrm{a}$ **and 1.**
Therefore, area of the triangle bounded by tangent and the axes $\frac{1}{2}\left|-\frac{1}{\mathrm{a}} \cdot 1\right| = \frac{1}{2|\mathrm{a}|}$

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It is minimum when $\mathrm{a} = 2$

Therefore, minimum area $= 1/4$

Q.15 Answer:

$2/3$

Solution:

$\lambda x - y + 2(1+\lambda) = 0 \Rightarrow \lambda(x+2) - (y-2) = 0$

This line passes through $(-2, 2)$.

$\mu \mathrm{x} - \mathrm{y} + 2(1-\mu) = 0 \Rightarrow \mu(\mathrm{x}-2) - (\mathrm{y}-2) = 0$

This line passes through $(2, 2)$.

Clearly these represents the foci of the ellipse.

So, $2ae = 4$

The circle $x^2 + y^2 - 4y - 5 = 0 \Rightarrow x^2 + (y-2)^2 = 9$ **represents auxiliary circle.**

Thus $a^2 = 9 \Rightarrow e = 2/3$ **and** $b^2 = 5$.

Required area $= \frac{1}{2}(2\mathrm{ae})(\mathrm{b}) = 3\frac{2}{3}\sqrt{5} = 2\sqrt{5}$

Q.16 Answer:

$2\sqrt{5}$

Solution:

$\lambda x - y + 2(1+\lambda) = 0 \Rightarrow \lambda(x+2) - (y-2) = 0$

This line passes through $(-2, 2)$.

$\mu \mathrm{x} - \mathrm{y} + 2(1-\mu) = 0 \Rightarrow \mu(\mathrm{x}-2) - (\mathrm{y}-2) = 0$

This line passes through $(2, 2)$.

Clearly these represents the foci of the ellipse.

So, $2\mathrm{ae} = 4$

The circle $x^2 + y^2 - 4y - 5 = 0 \Rightarrow x^2 + (y-2)^2 = 9$ **represents auxiliary circle.**

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Required area $= \frac{1}{2}(2\mathrm{ae})(\mathrm{b}) = 3\frac{2}{3}\sqrt{5} = 2\sqrt{5}$

Q.17 Answer:

None of these

Solution:

$\lambda x - y + 2(1+\lambda) = 0 \Rightarrow \lambda(x+2) - (y-2) = 0$

This line passes through $(-2, 2)$.

$\mu \mathrm{x} - \mathrm{y} + 2(1-\mu) = 0 \Rightarrow \mu(\mathrm{x}-2) - (\mathrm{y}-2) = 0$

This line passes through $(2, 2)$.

Clearly these represents the foci of the ellipse.

So, $2\mathrm{ae} = 4$

The circle $x^2 + y^2 - 4y - 5 = 0 \Rightarrow x^2 + (y-2)^2 = 9$ **represents auxiliary circle.**

Thus $a^2 = 9 \Rightarrow e = 2/3$ **and** $b^2 = 5$.

Required area $= \frac{1}{2}(2\mathrm{ae})(\mathrm{b}) = 3\frac{2}{3}\sqrt{5} = 2\sqrt{5}$

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$\Rightarrow \frac{x^2}{9}+\frac{(y-2)^2}{5}=1$

equation of ellipse.

$P(3\cos\theta, 2+5\sin\theta)$ **is any point on ellipse light from focus** $S'(-2,2)$ **touches** P **and gets reflected and passes through** $(2,2)$

$\Rightarrow \mathrm{S'P+SP=2a}$ **is always contact irrespective of the location** P.

Q.18 Answer:

A-q,s ; B-r ; C-p,q ; D-q,r

Solution:

(A) - q,s, (B) - r. (C) - p,q, (D) - q,r.

(A) Tangent to parabola having slope m **is** $\mathrm{ty=x+t^2}$**, it passes through point** $(2,3)$ **then** $3\mathrm{t}=2+\mathrm{t}^2 \Rightarrow \mathrm{t}=1$ **or** $t=2$

$\Rightarrow$ **point of contact** $\left(t^2+2t\right)\equiv(1,2)$ **or** $(4,4)$

(B) Let point on the circle be $\mathrm{P}\left(\mathrm{x}_1,\mathrm{y}_1\right)$ **then chord of contact of parabola w.r.t.** P **is** $yy_1=2\left(x+x_1\right)$.

Comparing with $\mathrm{y}=2(\mathrm{x}-2)$**, we have** $\mathrm{y}_1=1$ **and** $\mathrm{x}_1=2$**, which is also satisfy the circle.**

(C) Point Q on the parabola Q $\left(\mathrm{t}^2,2\mathrm{t}\right)$

Now area of triangle OPQ is $\frac{1}{2}\begin{vmatrix}0 & 0\\ 4 & -4\\ \mathrm{t}^2 & 2\mathrm{t}\\ 0 & 0\end{vmatrix}=6$

$\Rightarrow 8\mathrm{t}+4\mathrm{t}^2=\pm 12$

For $t^2+2t-3=0, \quad (t-1)(t+3)=0$**, then** $t=1$ **or** $t=-3$.

Then point Q **are** $(1,2)$ **or** $(9,-6)$

(D) Points $(1,2)$ **and** $(-2,1)$ **satisfy both the curves.**

Q.19 Answer:

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Solution:

$\left(\frac{t_2-t_1}{t_2^2-t_1^2}\right)\left(\frac{t_1-2}{t_1^2-4}\right)=-1$

Put $t_1\neq t_2$ **and** $t_1\neq 2$

$\Rightarrow \frac{1}{(\mathrm{t}_1+\mathrm{t}_2)}\frac{1}{(\mathrm{t}_1+2)}=-1 \Rightarrow \mathrm{t}_1^2+\mathrm{t}_1\mathrm{t}_2+2\mathrm{t}_1+2\mathrm{t}_2+1=0$

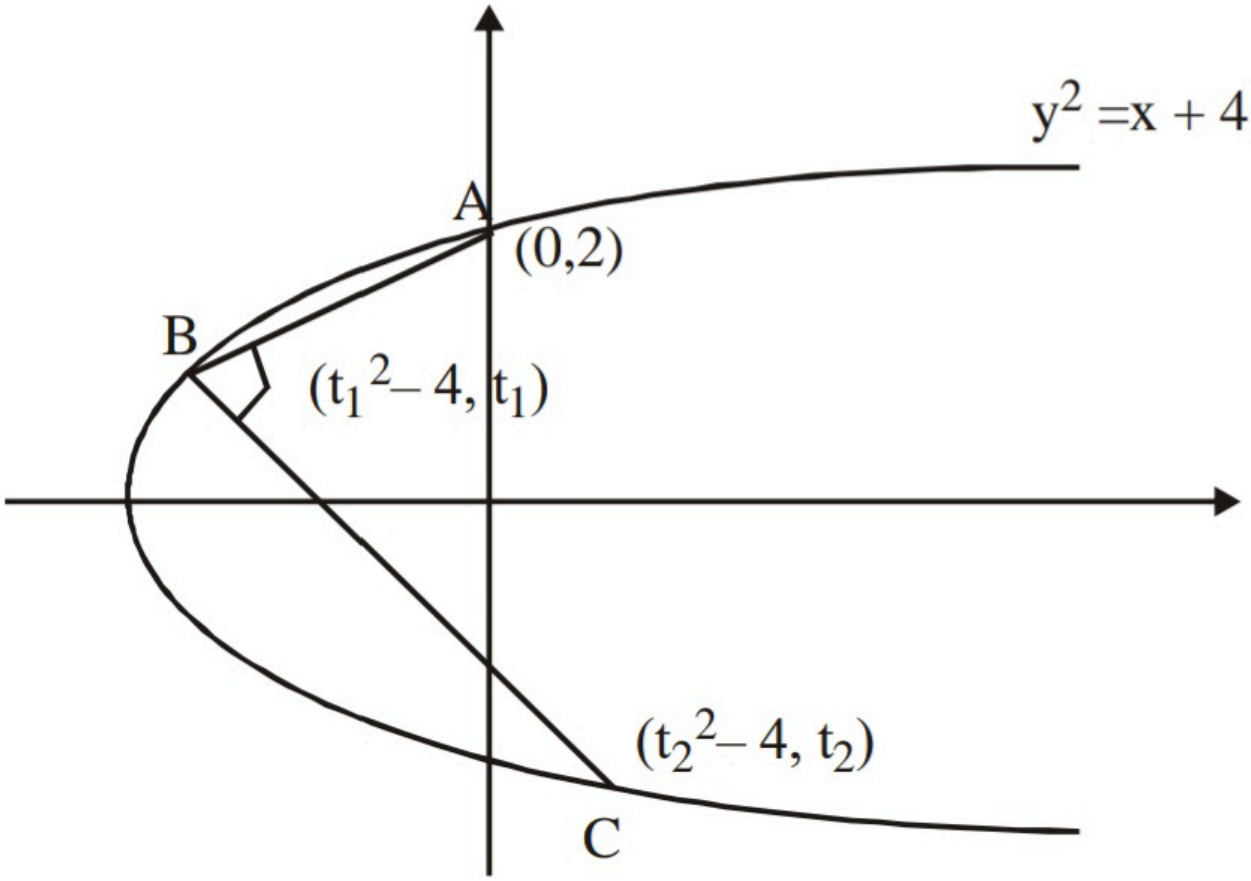
Quadratic in t_1 **should have real root**

$\Rightarrow \left(\mathrm{t}_2+2\right)^2-4\left(2\mathrm{t}_2+1\right)\geq 0$

$\Rightarrow \mathrm{t}_2^2-4\mathrm{t}_2\geq 0$

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$\Rightarrow \quad t_2 \leftarrow (-\infty, 0] \cup [4, \infty)$



There least positive value of t_2 is 4.

Q.20 Answer:

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Solution:

$\mathrm{BC} = \frac{\left|2t - t^2 - 17\right|}{\sqrt{5}} = a$

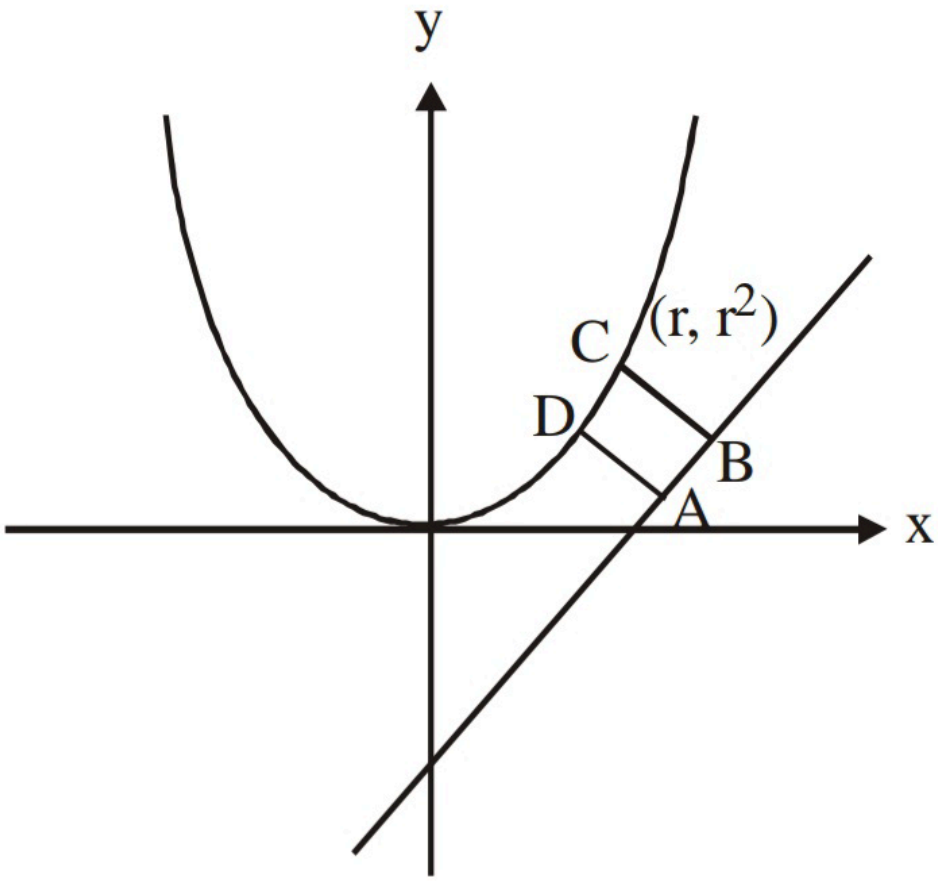
$\Rightarrow t^2 - 2t + 17 - = \pm\sqrt{5}$

If $C\left(t_1, t_1^2\right), D\left(t_2, t_2^2\right)$. then t_1, t_2 are roots of eq. (1) and $CD = a$

$\Rightarrow t_1 + t_2 = 2, t_1 t_2 = 17 - \sqrt{5}a$

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$\Rightarrow \quad a^2 = (t_1 - t_2)^2 + \left(t_1^2 - t_2^2\right)^2$



$a^2 = 4 - 4(17 - \sqrt{5}a) + 4(4 - 4(17 - \sqrt{5}a))$

$a^2 = 20 - 20(17) + 20\sqrt{5}a \Rightarrow a^2 - 20\sqrt{5}a + 320$

$\Rightarrow \quad a = 16\sqrt{5}$ or $a = 4\sqrt{5}$

Minimum area is when $a = 4\sqrt{5} \Rightarrow a^2 = 80 \Rightarrow A/16 = 5$

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